Normalization Report

Hospital Management System

Project Title : Hospital Management System

Domain : Healthcare

Database : HospitalManagementSystem

Tool : Microsoft SQL Server (SSMS)

Submitted by : Sundaravela A

---

1. Introduction

Normalization is the process of organizing a relational database to reduce data redundancy and improve data integrity. This report documents the normalization process applied to the Hospital Management System database up to Third Normal Form (3NF).

Table	Purpose
SPECIALIZATION	Stores medical specializations and departments
DOCTOR	Stores doctor details linked to specialization
PATIENT	Stores patient personal and contact details
APPOINTMENT	Stores appointment bookings between patient and doctor
TREATMENT	Stores diagnosis and prescription after appointment
BILLING	Stores bill and payment details per appointment
AUDIT_LOG	Tracks every change made to billing records

2. First Normal Form (1NF)

Rule

- Every column must contain atomic (single) values
- No repeating groups or multiple values in one cell
- Each row must be uniquely identifiable

```sql
-- Non 1NF: Multiple doctors stored in one cell (violates atomicity)
CREATE TABLE Patient_Non1NF (
    PatientID   INT,
    PatientName VARCHAR(100),
    Phone       VARCHAR(50),
    Doctors     VARCHAR(255)
);
```

```sql
 -- 1NF: One value per cell, each row is unique
CREATE TABLE Patient_1NF (
   PatientID   INT,
   PatientName VARCHAR(100),
   Phone       VARCHAR(50),
   Doctor      VARCHAR(100)
);
```

```sql
SELECT * FROM Patient_Non1NF;
SELECT * FROM Patient_1NF;
```

Results | Messages

	PatientID	PatientName	Phone	Doctors
1	1	James Carter	9876543210	Dr. John Smith, Dr. Sarah Connor
2	2	Emily Davis	9845612345	Dr. Robert Miles
3	3	Michael Brown	9823456789	Dr. Sarah Connor, Dr. John Smith, Dr. Alan Grant
4	4	Sophia Wilson	9812345678	Dr. Alan Grant, Dr. Robert Miles

	PatientID	PatientName	Phone	Doctor
1	1	James Carter	9876543210	Dr. John Smith
2	1	James Carter	9876543210	Dr. Sarah Connor
3	2	Emily Davis	9845612345	Dr. Robert Miles
4	3	Michael Brown	9823456789	Dr. Sarah Connor
5	3	Michael Brown	9823456789	Dr. John Smith

1NF Final Summary

TABLE	1NF STATUS	REASON
SPECIALIZATION	✅ Pass	All columns atomic
DOCTOR	✅ Pass	Name split into First & Last
PATIENT	✅ Pass	Name split, Gender as CHAR(1)
APPOINTMENT	✅ Pass	One patient, one doctor per row
TREATMENT	✅ Pass	One diagnosis per appointment
BILLING	✅ Pass	One payment method per bill
AUDIT_LOG	✅ Pass	One action per log entry

3. Second Normal Form (2NF)

Rule

- Must satisfy 1NF
- Every non-key column must depend on the WHOLE primary key
- No partial dependencies allowed

```
-- Non 2NF: DoctorName and DoctorPhone partially depend
-- on DoctorID only, not on full composite key
-- Problem: DoctorName repeats for every appointment of same doctor
CREATE TABLE Appointment_Non2NF (
   AppointmentID   INT,
   DoctorID       INT,
   PatientID      INT,
   DoctorName     VARCHAR(100),
   DoctorPhone    VARCHAR(15),
   AppointmentDate DATETIME,
   Status         VARCHAR(20)
);
```

```
-- 2NF Fix: Breaking Appointment_Non2NF into two tables
-- Table 1: DOCTOR_2NF (DoctorName depends only on DoctorID)
-- Table 2: APPOINTMENT_2NF (Only appointment specific columns)

-- ────────────────────────────────────────
-- BROKEN TABLE 1: Doctor
-- ────────────────────────────────────────
CREATE TABLE Doctor_2NF (
   DoctorID    INT PRIMARY KEY,
   DoctorName  VARCHAR(100),
   DoctorPhone VARCHAR(15)
);

-- ────────────────────────────────────────
-- BROKEN TABLE 2: Appointment
-- ────────────────────────────────────────
CREATE TABLE Appointment_2NF (
   AppointmentID   INT PRIMARY KEY,
   DoctorID        INT FOREIGN KEY REFERENCES Doctor_2NF(DoctorID),
   PatientID       INT,
   AppointmentDate DATETIME,
   Status          VARCHAR(20)
);
```

```sql
-- Join to prove relationship is maintained
SELECT
    A.AppointmentID,
    D.DoctorName,
    D.DoctorPhone,
    A.PatientID,
    A.AppointmentDate,
    A.Status
FROM Appointment_2NF A
INNER JOIN Doctor_2NF D ON A.DoctorID = D.DoctorID;
```

Results | Messages

	AppointmentID	DoctorName	DoctorPhone	PatientID	AppointmentDate	Status
1	1	Dr. John Smith	9876501234	201	2024-01-10 10:00:00.000	Completed
2	2	Dr. John Smith	9876501234	202	2024-01-11 11:00:00.000	Completed
3	3	Dr. Sarah Connor	9876502345	203	2024-01-12 09:00:00.000	Scheduled
4	4	Dr. Sarah Connor	9876502345	204	2024-01-13 14:00:00.000	Completed
5	5	Dr. Robert Miles	9876503456	201	2024-01-14 15:00:00.000	Cancelled

2NF Final Summary

TABLE	PRIMARY KEY	PARTIAL DEPENDENCY	2NF STATUS
SPECIALIZATION	SpecializationID	✅ None	✅ Pass
DOCTOR	DoctorID	✅ None	✅ Pass
PATIENT	PatientID	✅ None	✅ Pass
APPOINTMENT	AppointmentID	✅ None	✅ Pass
TREATMENT	TreatmentID	✅ None	✅ Pass
BILLING	BillID	✅ None	✅ Pass
AUDIT_LOG	LogID	✅ None	✅ Pass

4. Third Normal Form (3NF)

Rule

- Must satisfy 2NF
- No transitive dependencies
- Every non-key column must depend directly on the primary key only

```
-- Non 3NF: Transitive dependency present
-- DoctorID → SpecName → Department
-- Department does not directly depend on DoctorID
CREATE TABLE Doctor_Non3NF (
    DoctorID   INT PRIMARY KEY,
    DoctorName VARCHAR(100),
    SpecName   VARCHAR(100),
    Department VARCHAR(100)
);
```

```
-- 3NF Fix: Breaking Doctor_Non3NF into two tables
-- Table 1: SPECIALIZATION_3NF (Department depends on SpecName not DoctorID)
-- Table 2: DOCTOR_3NF (Only doctor specific columns + FK reference)

-- ────────────────────────────────────────────
-- BROKEN TABLE 1: Specialization
-- ────────────────────────────────────────────
CREATE TABLE Specialization_3NF (
    SpecializationID INT PRIMARY KEY,
    SpecName         VARCHAR(100),
    Department       VARCHAR(100)
);

-- ────────────────────────────────────────────
-- BROKEN TABLE 2: Doctor
-- ────────────────────────────────────────────
CREATE TABLE Doctor_3NF (
    DoctorID         INT PRIMARY KEY,
    DoctorName       VARCHAR(100),
    SpecializationID INT FOREIGN KEY REFERENCES Specialization_3NF(SpecializationID)
);
```

```sql
-- Join to prove relationship is maintained
SELECT
    D.DoctorID,
    D.DoctorName,
    S.SpecName,
    S.Department
FROM Doctor_3NF D
INNER JOIN Specialization_3NF S ON D.SpecializationID = S.SpecializationID;
```

Results | Messages

	DoctorID	DoctorName	SpecName	Department
1	101	Dr. John Smith	Cardiology	Heart & Vascular
2	102	Dr. Sarah Connor	Cardiology	Heart & Vascular
3	103	Dr. Robert Miles	Neurology	Brain & Spine
4	104	Dr. Alan Grant	Neurology	Brain & Spine
5	105	Dr. Laura Hayes	Orthopedics	Bone & Joint

3NF Final Summary

TABLE	TRANSITIVE DEPENDENCY	HOW RESOLVED	3NF STATUS
SPECIALIZATION	✅ None	SpecName & Department depend directly on SpecializationID	✅ Pass
DOCTOR	✅ None	SpecName & Department moved to SPECIALIZATION table	✅ Pass
PATIENT	✅ None	All columns depend directly on PatientID	✅ Pass
APPOINTMENT	✅ None	Patient & Doctor details stored in their own tables	✅ Pass
TREATMENT	✅ None	All columns depend directly on TreatmentID	✅ Pass
BILLING	✅ None	All columns depend directly on BillID	✅ Pass
AUDIT_LOG	✅ None	All columns depend directly on LogID	✅ Pass

5. Overall Normalization Summary

NORMAL FORM	RULE	ALL 7 TABLES
1NF	Atomic values, no repeating groups	✅ All Pass
2NF	No partial dependency	✅ All Pass
3NF	No transitive dependency	✅ All Pass

6. Key Design Decisions

DECISION	REASON
SPECIALIZATION created as separate table	Removes transitive dependency — Department depends on SpecName not DoctorID
Name split into FirstName and LastName	Ensures atomicity — satisfies 1NF
Gender stored as CHAR(1)	Single atomic value — satisfies 1NF
All tables use single column PK	Eliminates possibility of partial dependency — satisfies 2NF
FK references used instead of storing data	Prevents data duplication — satisfies 3NF
AUDIT_LOG as separate table	Ensures billing changes tracked without modifying BILLING table

7. Conclusion

The Hospital Management System database has been successfully normalized up to Third Normal Form (3NF).

- All 7 tables satisfy 1NF, 2NF and 3NF
- Data redundancy is completely eliminated
- Data integrity is maintained across all tables
- Foreign key relationships are properly defined
- SPECIALIZATION table introduced to resolve 3NF transitive dependency
- Single column primary keys used throughout to prevent partial dependencies
- The design is clean, scalable and fully ready for Sprint 2 implementation